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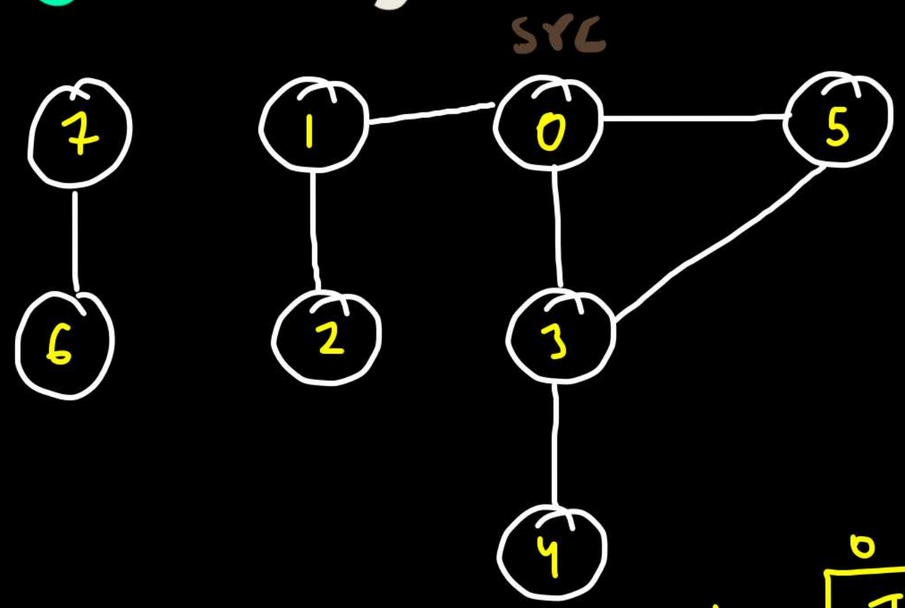
Ques: Cycle Detection in undirected graph



bfs ? ✓

dfs ? ✓

(5,0) (2,1) (4,3)

(0,-1) (1,0)

(3,0)

vis

0	1	2	3	4	5
T	T	T	T	T	T

Cycle caught

edges = {[7,6], [1,2], [1,0], [5,0], [0,3], [3,4]}

0 : 1, 5, 3
1 : 2, 0
2 : 1
3 : 0, 4
4 : 3
5 : 0
6 : 7
7 : 6

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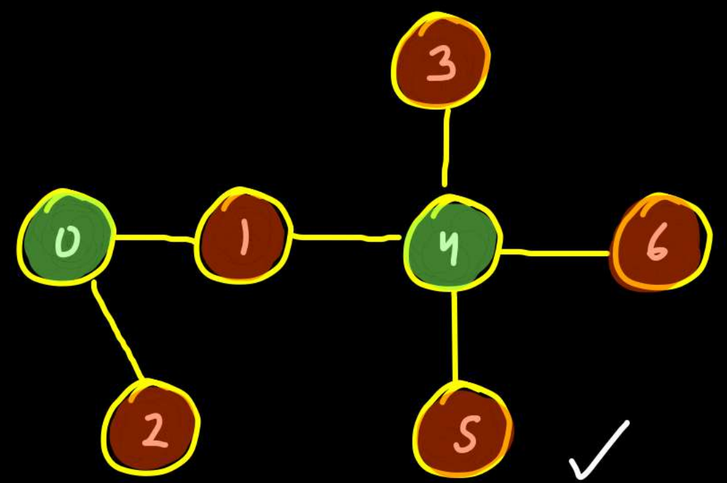
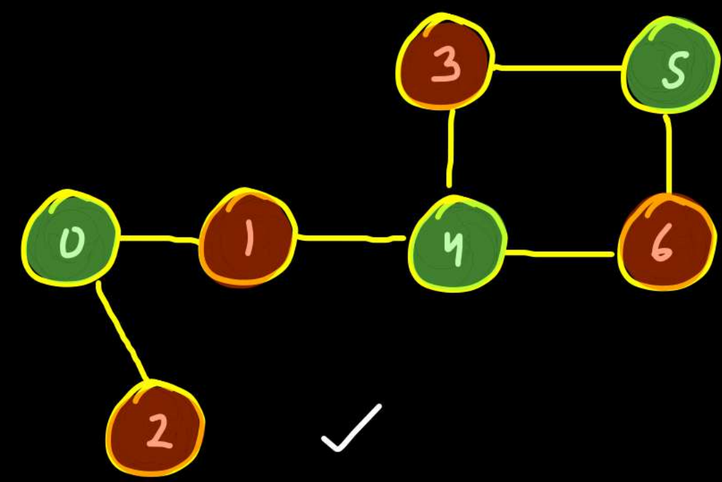
Ques: Bipartite Graph

cycle length = even

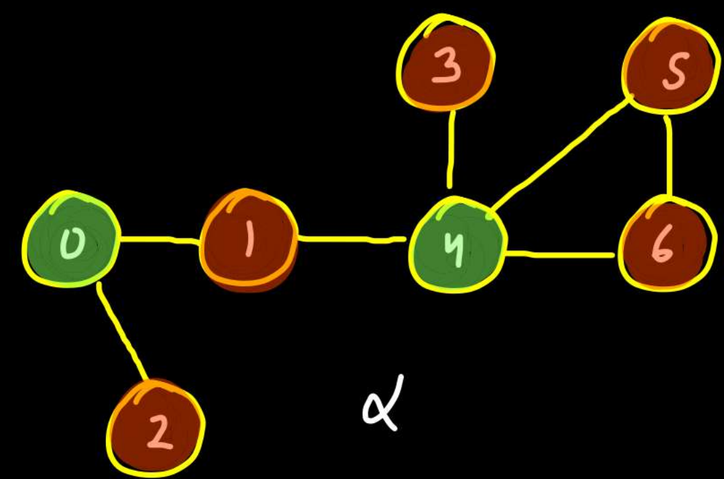


✓



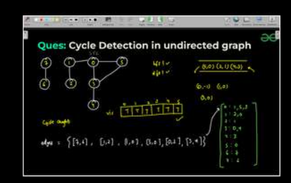
✓

if no cycle exists then graph is always bipartite



✗

cycle length is odd



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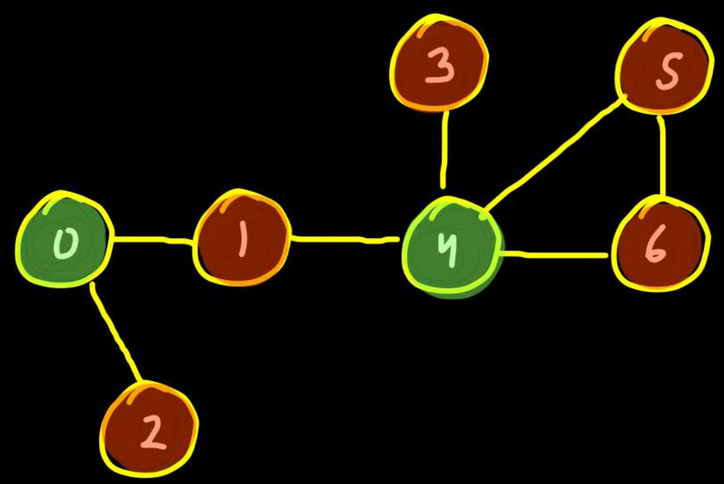
Ques: Bipartite Graph

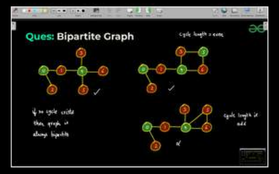


6

~~0~~ ~~2~~ ~~4~~ ~~2~~ 5 → bipartite

	0	1	2	3	4	5	6
vis	1	2	2	2	1	2	2

green = 1
red = 2



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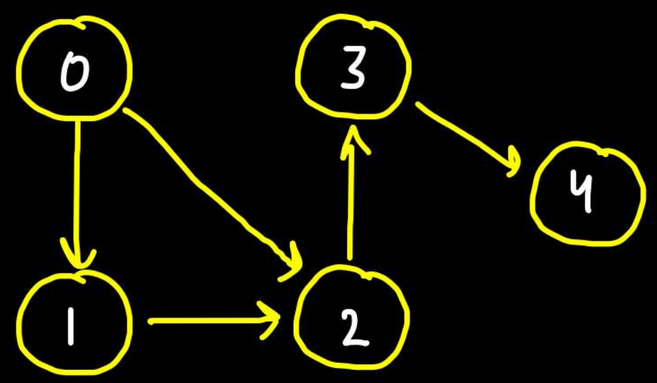
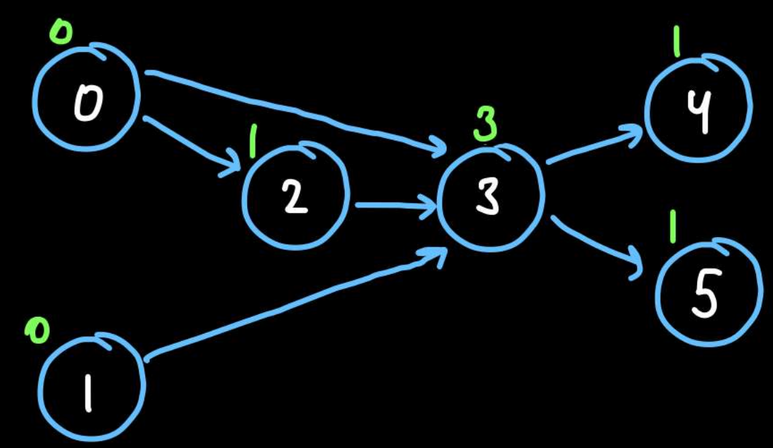
Directed Acyclic Graph (DAG)

edges = { [0,2], [2,3], [3,4], [1,3], [3,5] }





topo sort : 0 1 2 3 4

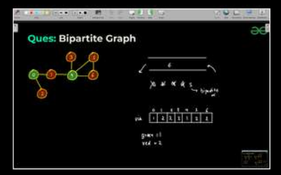
topo sort :

1 0 2 3 5 4 ✓

1 0 2 3 4 5 ✓

0 1 2 3 5 4 ✓

0 2 1 3 5 4 ✓



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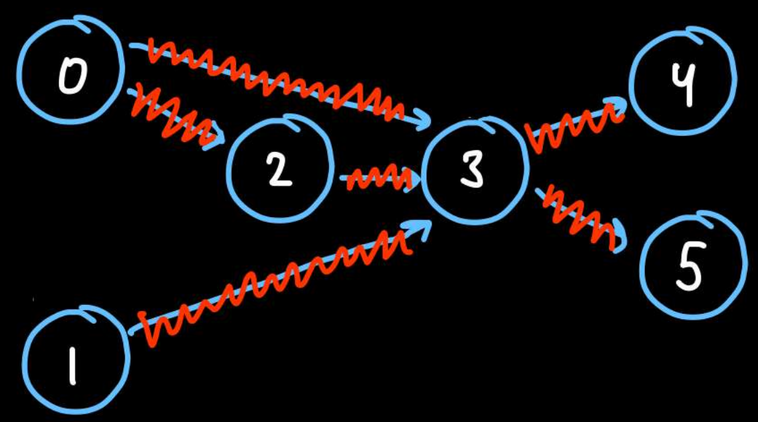
Topological Sort (Kahn's Algo) (BFS)

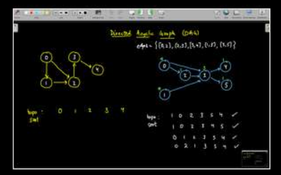
'q' me sirf 0 indegree wale nodes jaayenge

edges = { [0,2], [2,3], [3,4], [1,3], [3,5], [0,3] }



	0	1	2	3	4	5
in	0	0	0	0	0	0

ans = { 0 1 2 3 4 5 }



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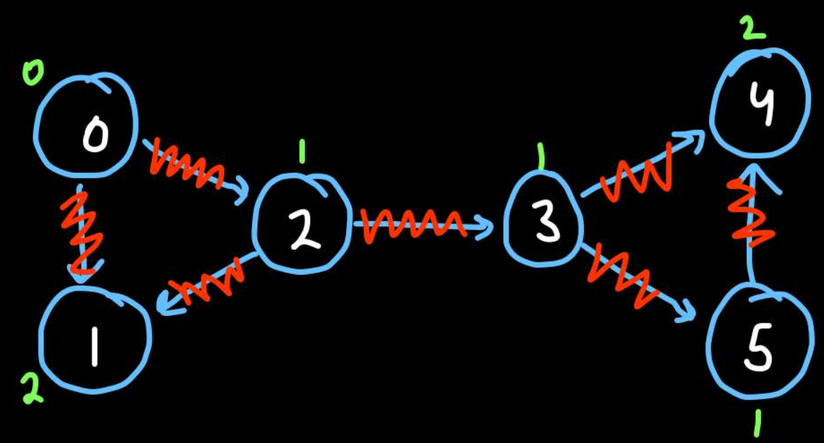
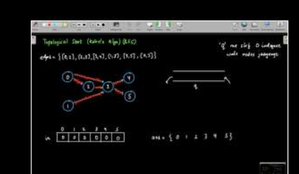
Ques: Topological Sort (DAG)



in

0	1	2	3	4	5
0	0	0	0	0	0

ans = {0, 2, 3, 1, 5, 4}



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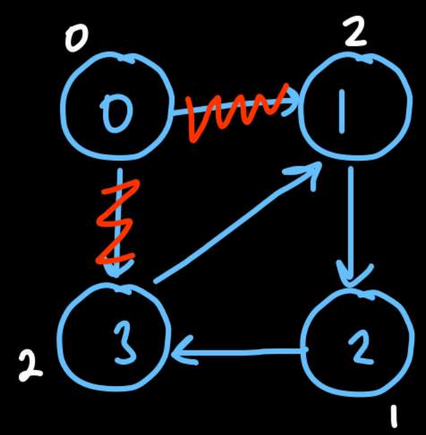
Ques: Cycle Detection in directed graph (Kahn's Algo)

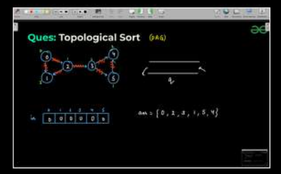


in

0	1	2	3
0	1	1	1

ans = { 0 }

length = 1
nodes = 4] true (cycle)



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HW Course Schedule - 1